

Company: BrowserStack

Job Profile: Software Developer: Backend

Offer Type: I only

Internship Stipend: 75k

CTC: upto 22 lpa

Location: Remote

Process Date: 23rd July OA & 31st July Interview

Placement Process:

1. OA round [92 appeared]

Platform : HackerEarth

Eligibility: CE and CSE, 7.5 CGPA cutoff

21 MCQ questions were asked in 50 min, primary HTML CSS JS SQL CCN questions were asked. The test was proctored and offline. The questions were easy to medium level. If you have built projects using JS it would be a piece of cake and the entire test could be completed in 30 mins max. Accuracy matters over here for clearing the cutoff.

2. Machine Coding Round [11 shortlisted 7 appeared]

It was conducted in Batches offline in the BrowserStack office in Oberoi Commerz 2. Due to the limited number of students they had confined a single batch for SPIT. The PS for the machine coding round is shared below:

This problem requires you to implement a log-watching solution (similar to the tail -f command in UNIX). However, the log file is hosted on a remote machine (same machine as your server code). The log file is in append-only mode.

You need to implement the following:

- A server-side program to monitor the given log file and stream updates as they happen using Socket.IO for real-time communication. This will run on the same machine as the log file. You can use any programming language for the server.
- A web-based client (accessible via a URL like <http://localhost/log>) that displays updates in the file in real-time, without requiring a page refresh. The page should load once and keep updating. On initial load, the user should see the last 10 lines of the file.

Problem Constraints:

- The server must push updates to clients in real-time using Socket.IO.
- The log file could be several GBs. How do you optimize retrieving the last 10 lines?
- The server should not retransmit the entire file every time; it should only send updates.

- The server must handle multiple clients simultaneously.
 - The web page should not stay in a loading state after the first load and should not reload thereafter.
 - You may not use off-the-shelf external libraries or tools to read the file or provide tail-like functionalities (except Socket.IO for real-time updates).
- You'll be evaluated on code quality, testability, modularity, and handling of corner cases.

The Machine coding round began at 10:30 AM and we were given roughly 2 hrs to build the backend server. You may find the solutions online, but an optimal solution that caters to the edge cases of the invigilator is preferred. You are not allowed to use any AI agents or LLMs or medium articles during this round, our invigilator was right beside us. You may refer to the documentation for development. You can interact with the invigilator during the round. Primary goal is to develop the solution using Socket.IO, I preferred NODE for the backend and socket implementation to display logs realtime with a confined Buffer to prevent parsing the logs.

The invigilator would test it on multiple edge cases, like real time parsing, multiple browser compatibility with realtime logs and order in which they appear or cases where '\n' appears while parsing logs. If it passes all the cases and you are able to explain your code really well then you would be moved forward to the next round.

3. **Director Of Engineering Round** [3 shortlisted]

For preparation you may go through medium or GFG articles.

Mine was an interesting one, I was the only one who had an online interview with the Senior Director in the office for 30 mins. He read my resume and given that I had worked with a corporation with exposure to multiple hacks and techstacks, was exempted from basic questions. I was asked about one of my favourite projects and what role did I play in that project. Then moved on to internship experiences, what tasks did I perform in Russell Investments as Summer Intern, what project did I work on. He asked if I had exposure to Prod and deployment which I explained pretty well. I was quickly moved to behavioral questions like how will I tackle Deadlines, How will I deal with my teammate who differs in an opinion. It was the most fun director round.

All the 3 received the offer.

Useful Tips:

1. Participate in Hacks
2. Get exposure (DSA helps but DEV is important :))
3. Pursue internships that help you learn and offcourse Paid one
4. Don't Stress

You can connect with me via linkedin <https://www.linkedin.com/in/paramgogia/> or if you are SPITian you can find me in TPO Announcements.

